Lecture Template

ADA Compliant

May 12, 2026

Architecture and Compliance

Lists/Tables/Figures

Math/Verbatim

Custom Frames and Blocks

Toggle Visibility Logic

UF Brand Center

Summary

Technical Requirements

Reference: Dr. Jason Nowell.

Critical Workflow Components:

- **Engine:** Use `LuaLaTeX` for its superior handling of OpenType math fonts, which is critical for MathML generation in order to populate math tags.
- **Kernel:** Use the Phase 3 Beta kernel (released Nov 1st, 2025, or later).
- **Preamble:** Use preamble and accessibility packages provided in this presentation and reference. For slides use (highly experimental) document class `ltx-talk`.
- **Validation:**
  - **Structural:** Use veraPDF (Industry Standard). Also recommend checker PAC 2026 with detailed report and built-in screen reader.
  - **Semantic:** Verify with screen readers (NVDA, JAWS, or VoiceOver).
- **Best Practices:** Prioritize structural simplicity to ensure tag-tree integrity.

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Lists

Ordered and unordered lists are automatically tagged for screen reader navigation. Lists in this template have been defaulted to have 10pt separation. Here is an example of a list with custom overlays:

- First item.

Lists

Ordered and unordered lists are automatically tagged for screen reader navigation. Lists in this template have been defaulted to have 10pt separation. Here is an example of a list with custom overlays:

- First item.
- Second item.

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- First item.
- Second item.
- Third item.

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- First item.
- Second item.
- Third item.
- Fourth item (while also displaying third).

Table

Tables require explicit header row identification for data association in screen readers. Use `\tagpdfsetup{table/header-rows={1}}`.

Known Limitation: If you encounter issues with regular captions, use the `makebox` workaround shown.

```
\makebox[0pt][c]{Table 1: Calculated Standard Errors}
```

Method	**Sample Size (n)**	**Standard Error**
Simple Random	100	0.050
Stratified	100	0.032
Cluster	100	0.075

Table 1: Calculated Standard Errors

Figures

- **Alt Text:** Mandatory for all non-artifact images.
- **Color Space:** Avoid CMYK (.eps). Use RGB (.png/.jpg) to ensure compliance with digital accessibility standards.

```
\includegraphics[width=0.5\textwidth,alt={University of Florida logo:
a blue rectangle containing the letters UF in orange and the words
UNIVERSITY of FLORIDA below.}]{Logo_RGB_Raster.png}
```

Figure 1: Official UF Signature Logo

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UF Brand Center

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Automatic MathML Support

An example of a slide containing math.

- **Sampling distribution:**
  Let $T(Y_1, Y_2, \dots, Y_n)$ be some *statistic*, i.e., some function of the data. Then, we call the probability distribution of $T(Y_1, Y_2, \dots, Y_n)$ the *sampling distribution*.

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- **Standard error:**
  The *standard error* of an estimator $\hat{\theta}$ is the standard deviation of the sampling distribution:

$$\mathrm{SE}\left(\hat{\theta}\right) = \sqrt{E\left[\left(\hat{\theta} - \theta\right)^2\right]}$$

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Block

This is a block, like a theorem. Background colors can cause issues, even if color scheme is appropriate.

Verbatim

Unlike beamer where we used:
`\begin{frame}[fragile]`

With `ltx-talk`, simply use
`\begin{frame*}...`
Note that this command cannot be nested within another command.

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Custom Blocks

Theory: Optional Title

This is a theorem

Custom Blocks

Example: Optional Title

This is an example.

Custom Blocks

Remark: Optional Title

This is a remark.

Example

This is an example that takes up the whole slide. A block is not sufficient. We create an example frame using:

```
\exampleon
\begin{frame}
...
\end{frame}
\exampleoff
```

Similarly we can completely remove the top banner using:

```
\banneroff
\begin{frame}
...
\end{frame}
\banneron
```

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Current Mode: **Solutions Visible**

In preamble comment out one:

\showsolutionstrue

\showsolutionsfalse

$$
\begin{aligned}
V(\hat{\mu}) &= V\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} V(Y_i) = \frac{\sigma^2}{n} \\
\implies \mathrm{SE}(\hat{\mu}) &= \frac{\sigma}{\sqrt{n}}
\end{aligned}
$$

Preamble Command Used

`\sol[space_when_false]{content when true}`
allows for a single-source workflow, generating an "Instructor" and "Student" version easily.

Overlay in `align*`

Use of `\onslide` within `align*` to present specific lines.

$$x = x$$

Overlay in `align*`

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$$x = x$$

$$y = y$$

Temporal Overlays

With the use of `\temporal`, can create faded overlays whereby a before, current, and after options can be chosen.

```
\NewDocumentCommand\fade{D<>{all}}{%
 \temporal<#1>{\color{black!70}}{\color{black}}{\color{black!10}}
}
```

1. First item.
2. Second item.
3. Third item.
4. Fourth item (while also displaying third).

Note: Unlike standard overlays, temporal overlays for lists retain visible bullets. The use of `\temporal` inside a list, the new kernel's tagging engine (tagpdf) forces the item bullet to be a protected "Label"

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Overlay Suppression and Other Info

To generate a static version for student notes, use:

```
\documentclass[mode=handout]{ltx-talk}
```

More information on overlay functions and environments in Section 9 of ltx-talk documentation.

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Colors and Typography

UF Brand Center has resources on appropriate use of:

- COLORS, including AA-Compliant Color Formulas for Screen Applications
- TYPOGRAPHY, including Extended Character Language Support and Accessibility

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- **Best Practices:** Still highly experimental, keep it simple.

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- **Preamble:** Appropriate calls are used in preamble. For presentations use `ltx-talk`.
- **Best Practices:** Still highly experimental, keep it simple.
- **Future:** Development is ongoing.